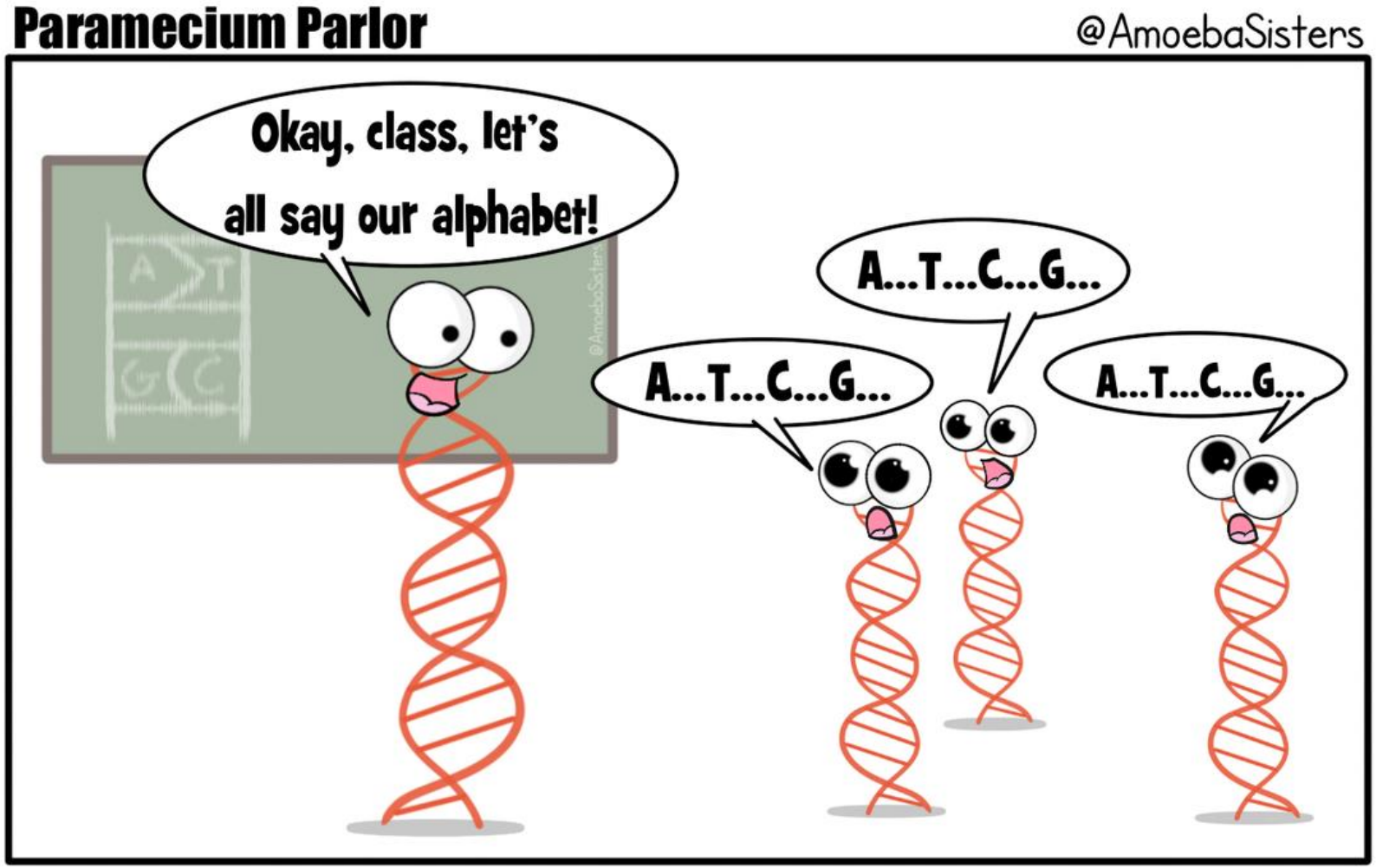
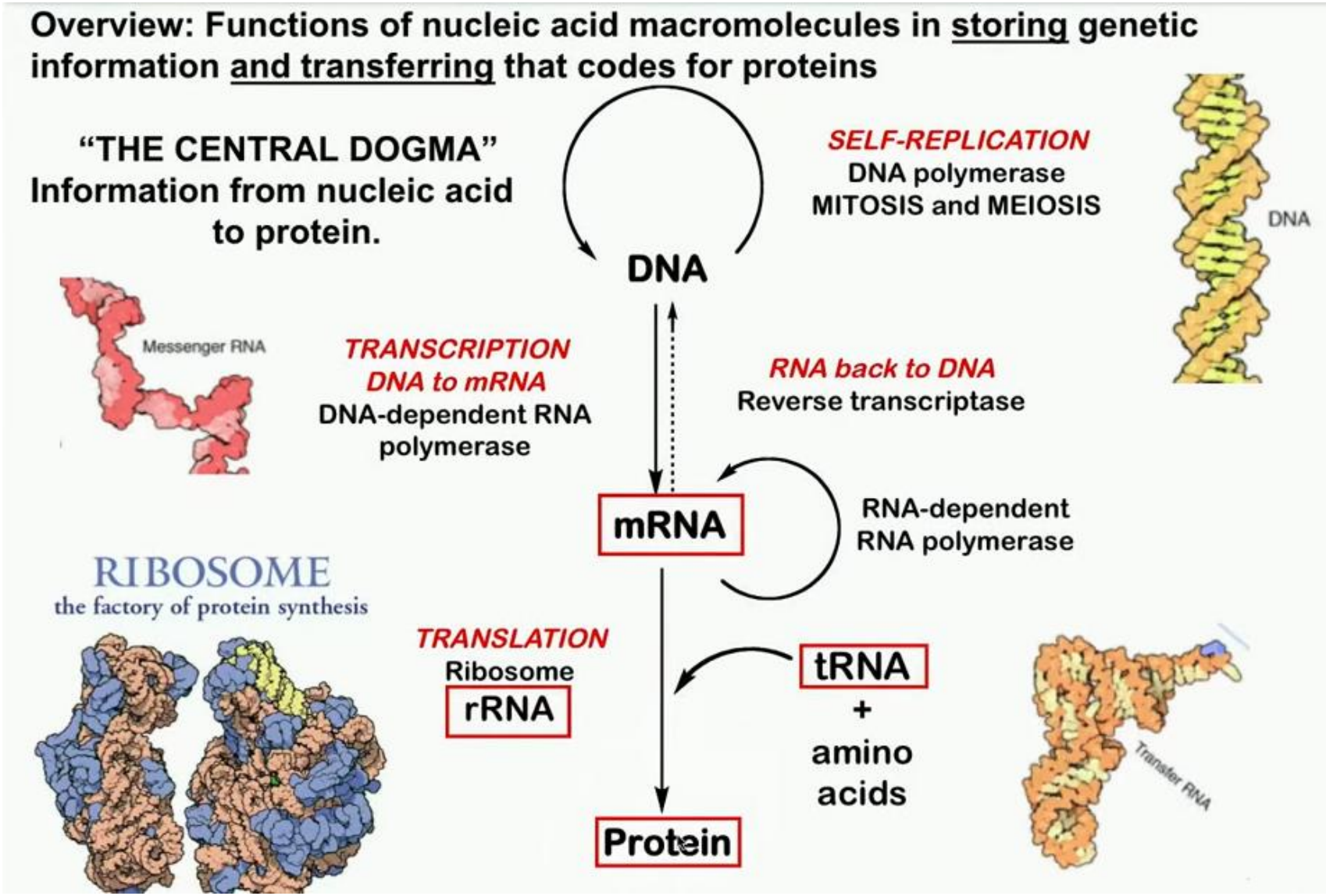
Replication

Lecture 10



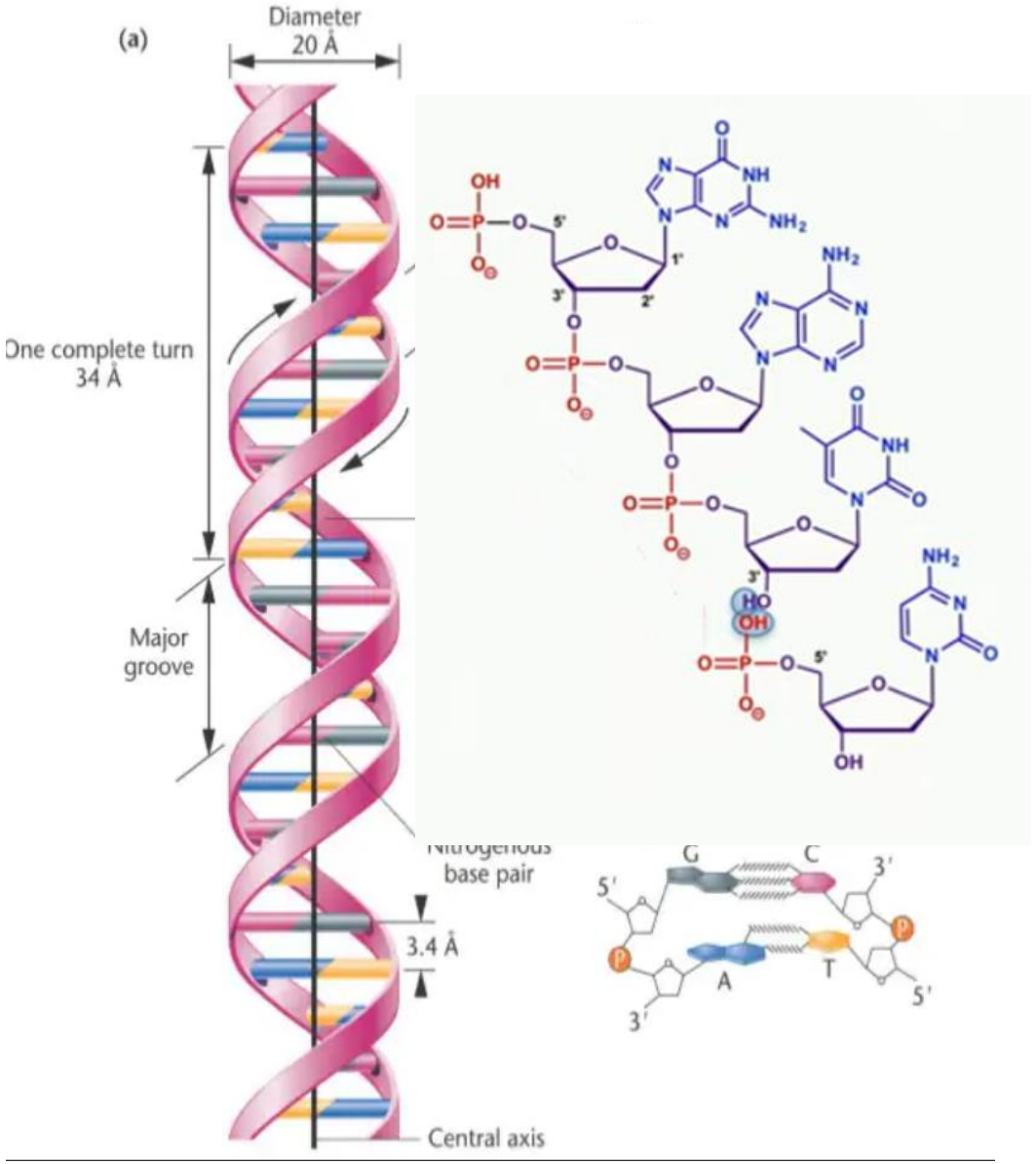
The molecules of life



`It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material'

Watson & Crick
Nature (1953)

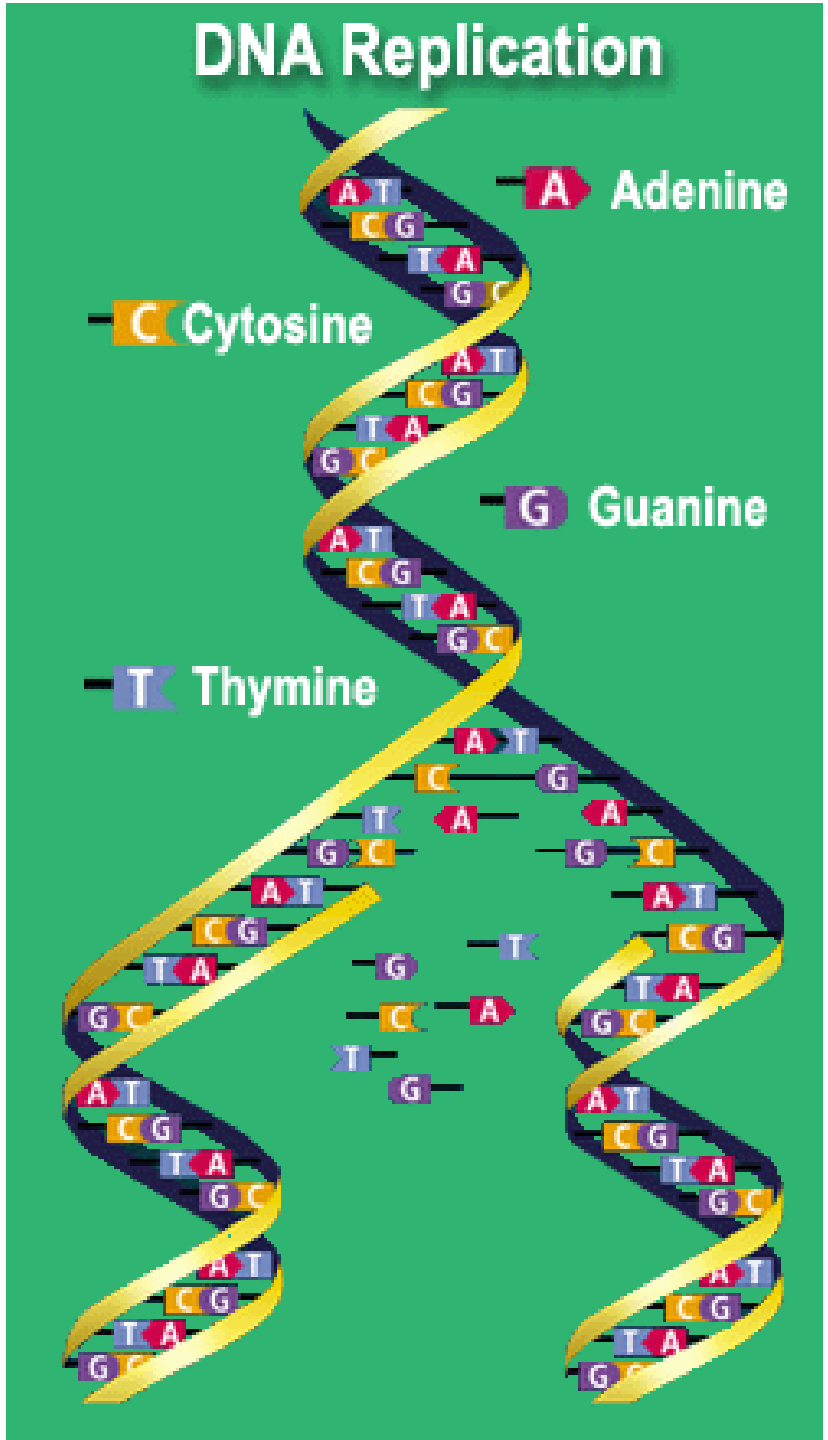
dsDNA flashback



Each dsDNA consist of:

1) A phosphate backbone
2) Purines binding with pyrimidine via hydrogen bonds (A-T and G-C)

- dsDNA are formed by antiparallel complementary strands that are joined together through base pairing – A with T and G with C.
- New nucleotides are always added to the 3'-End of the growing polymer
- dsDNA can be denatured (separated) and reannealed or hybridized multiple times.



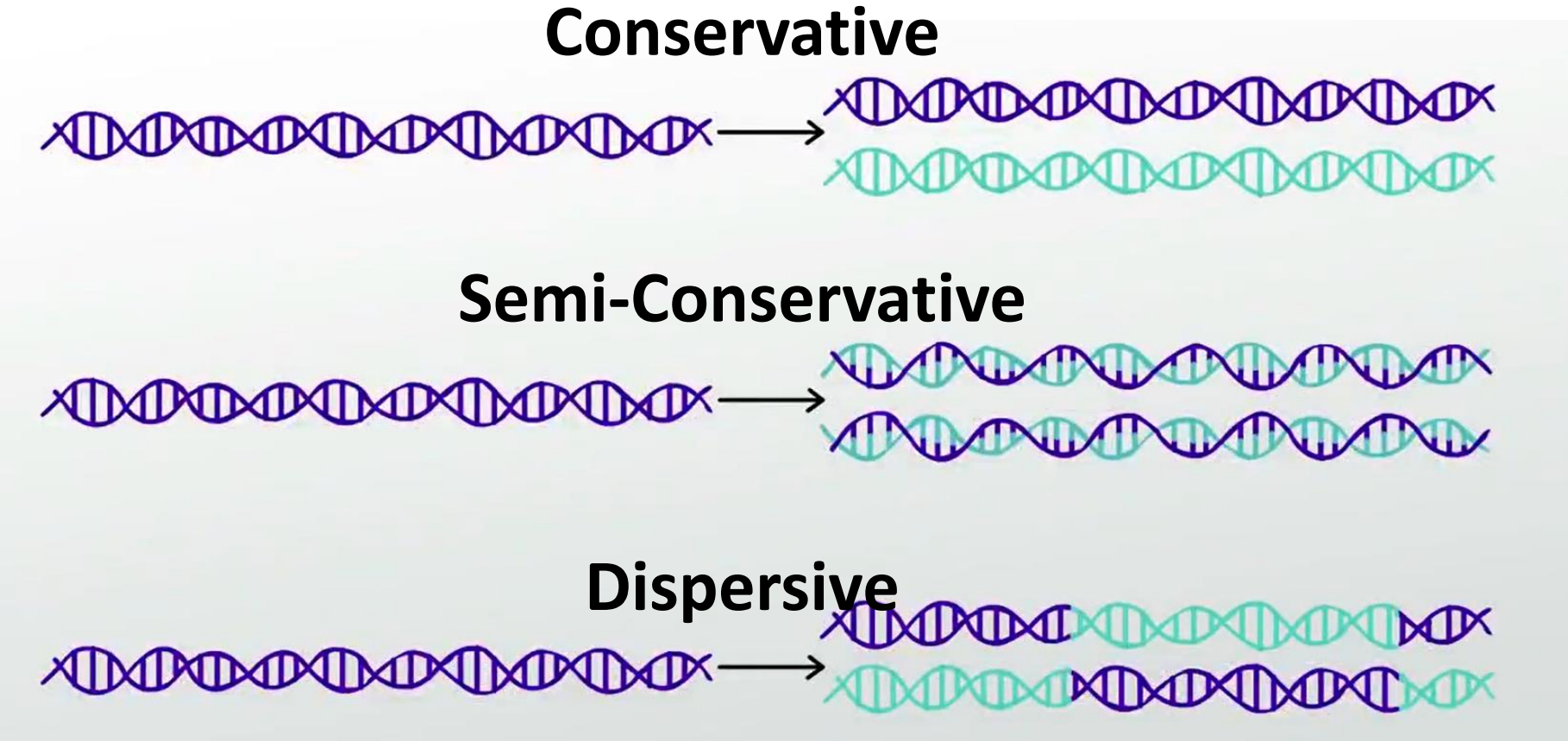
Step 1: The Double Helix is unzipped

Step 2: Each strand directs the synthesis of two new strands

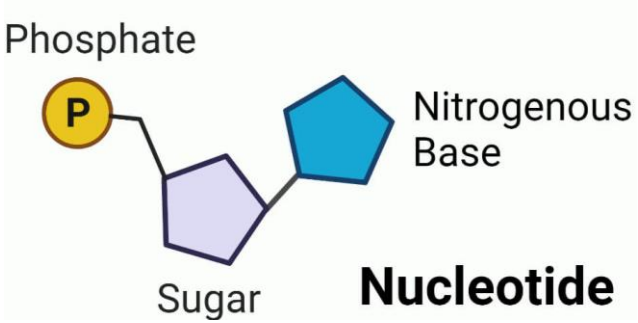
Step 3: The individual nucleotides are connected to form two new strands

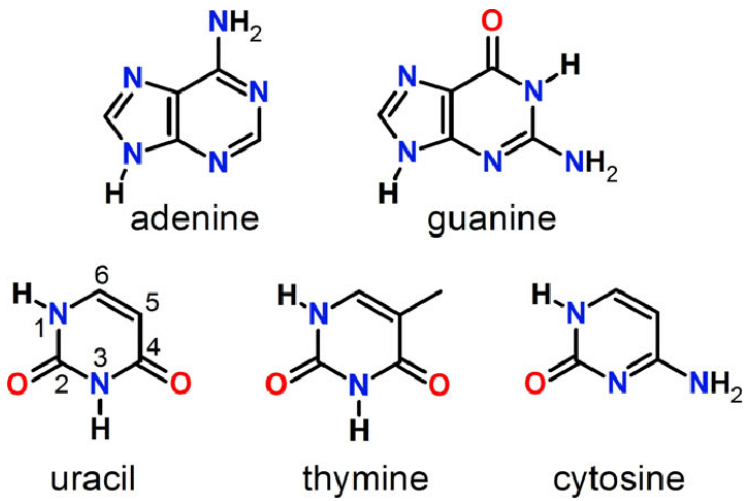
How is DNA replicated?

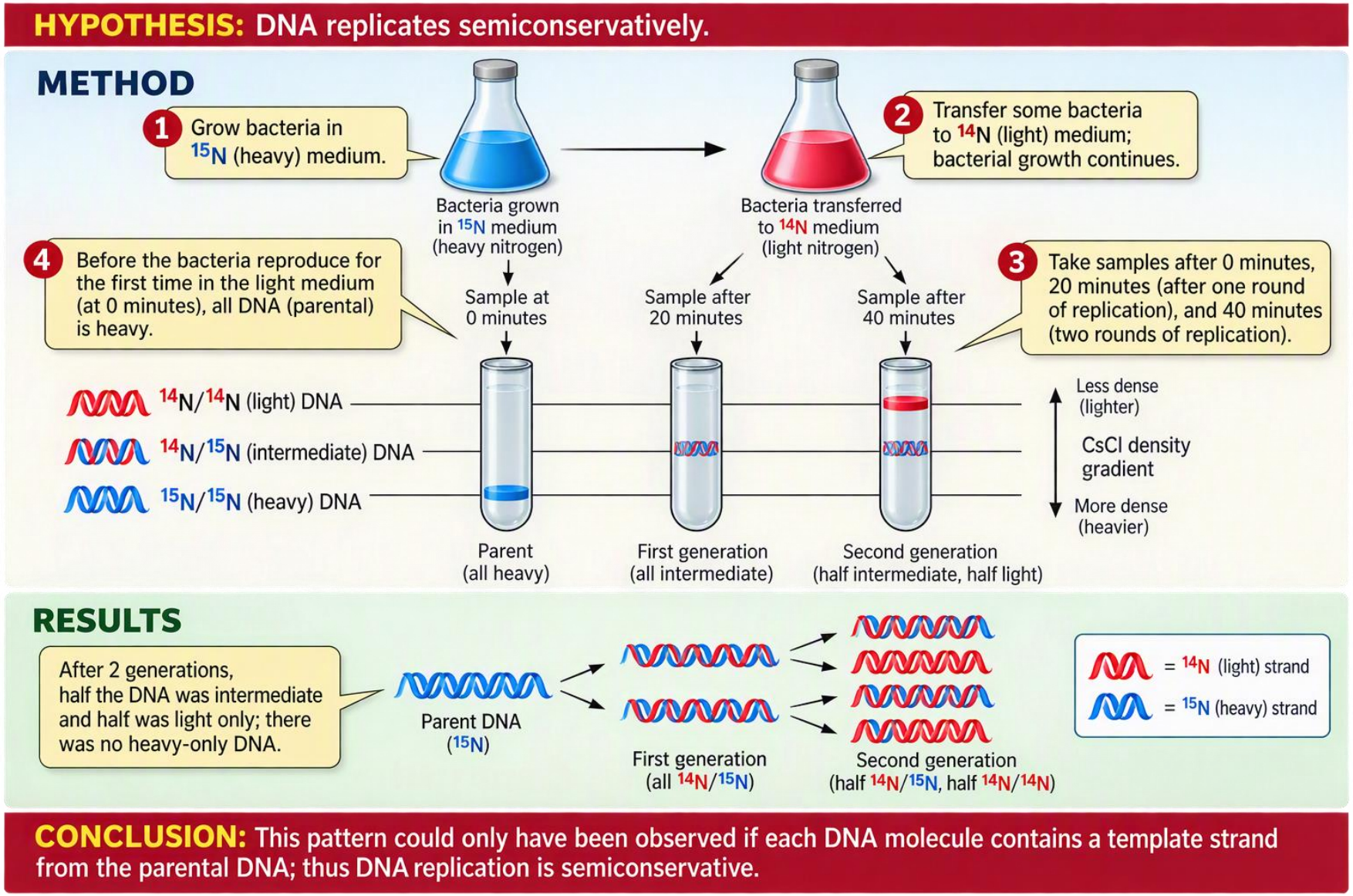
It was **known** that DNA is double helix, but still **unclear** how DNA was copied



Meselson and Stahl Experiment 1958



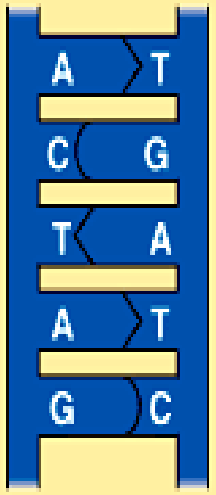




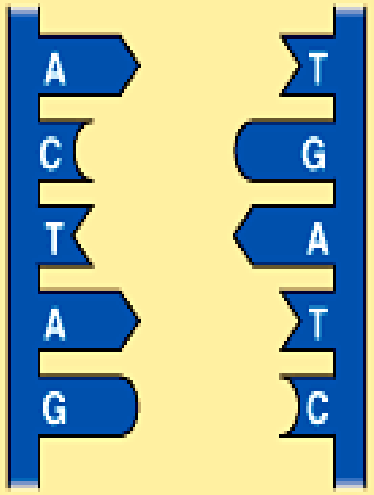
https://www.nature.com/scitable/content/the-meselson-stahl-experiment-18551/

https://www.pnas.org/doi/10.1073/pnas.44.7.671

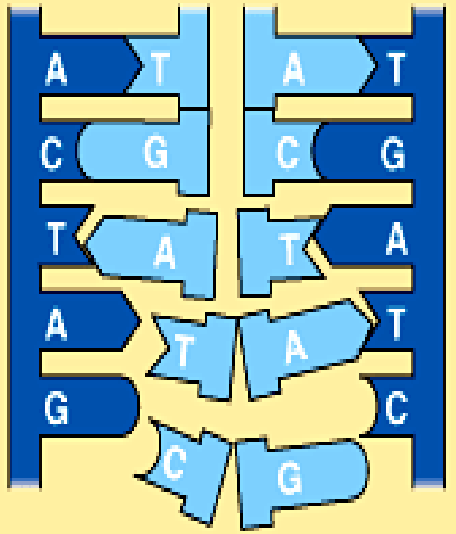
DNA Replication – Overview



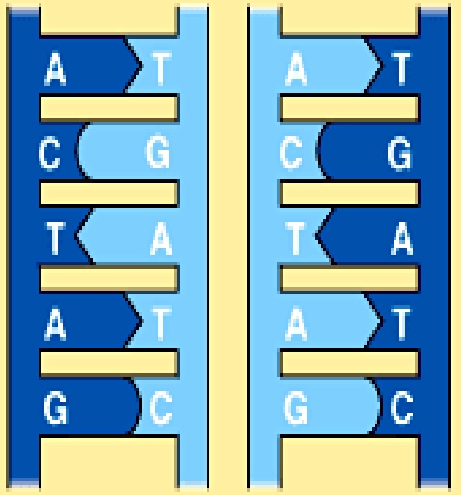
(a) The parent molecule has two complementary strands of DNA. Each base is paired by hydrogen bonding with its specific partner, A with T and G with C.



(b) The first step in replication is separation of the two DNA strands.



(c) Each parental strand now serves as a template that determines the order of nucleotides along a new complementary strand.



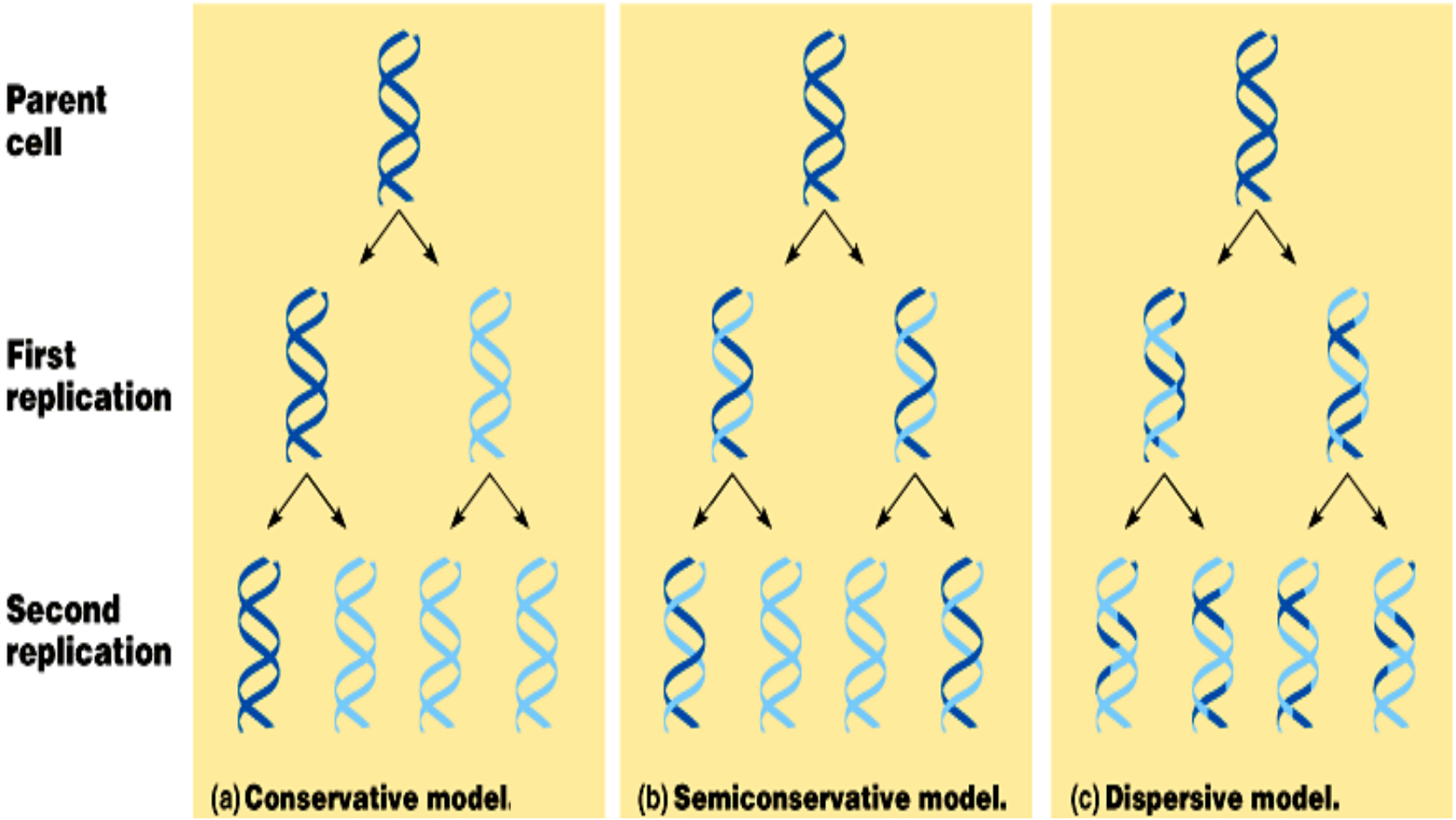
(d) The nucleotides are connected to form the sugar-phosphate backbones of the new strands. Each "daughter" DNA molecule consists of one parental strand and one new strand.

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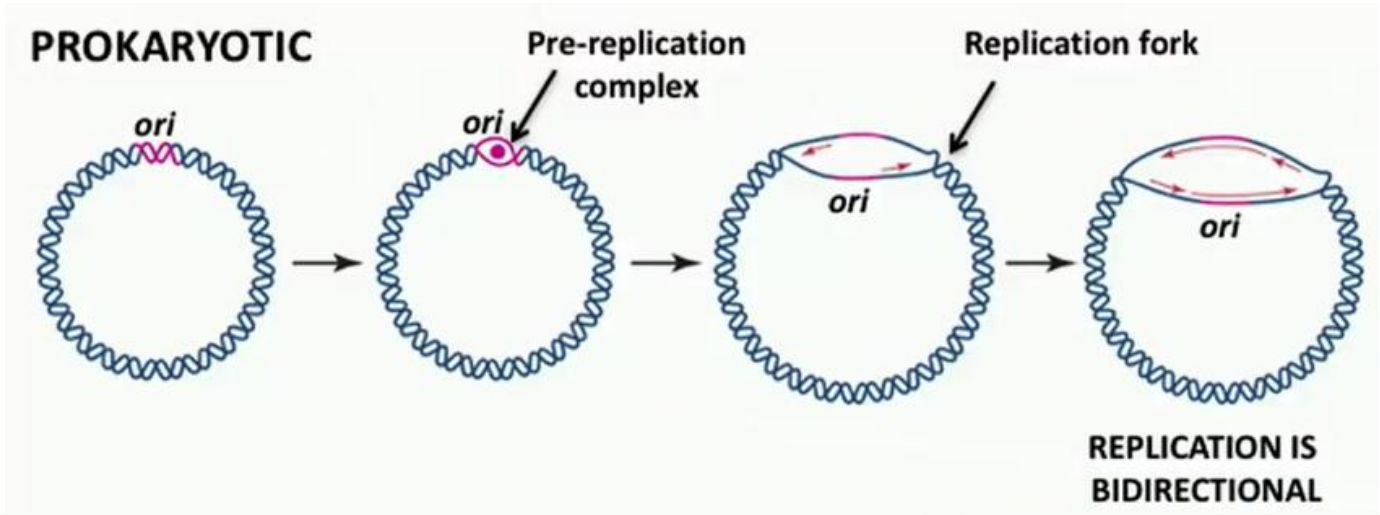
Semiconservative Replication

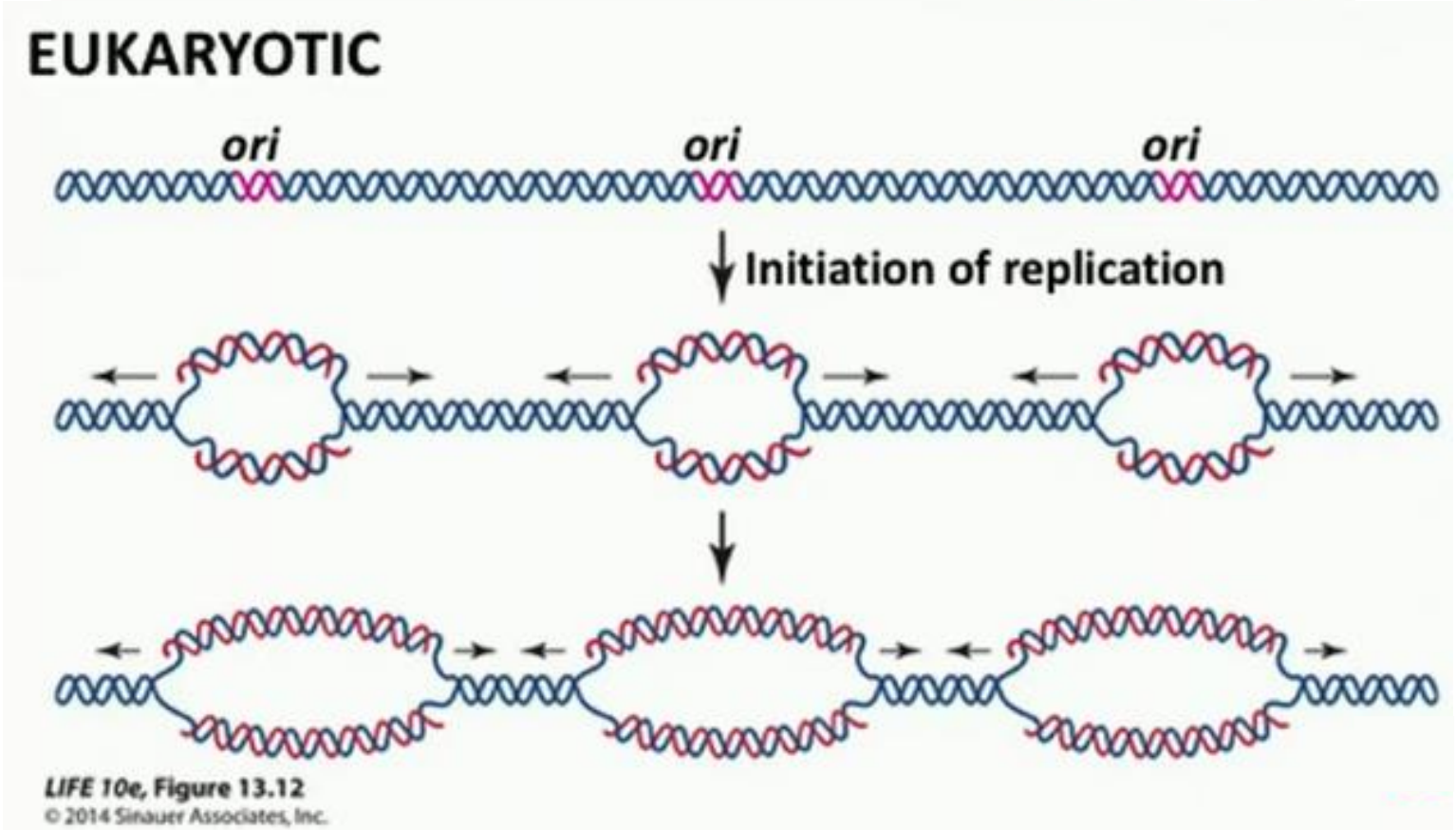
Process of DNA replication is called semiconservative replication

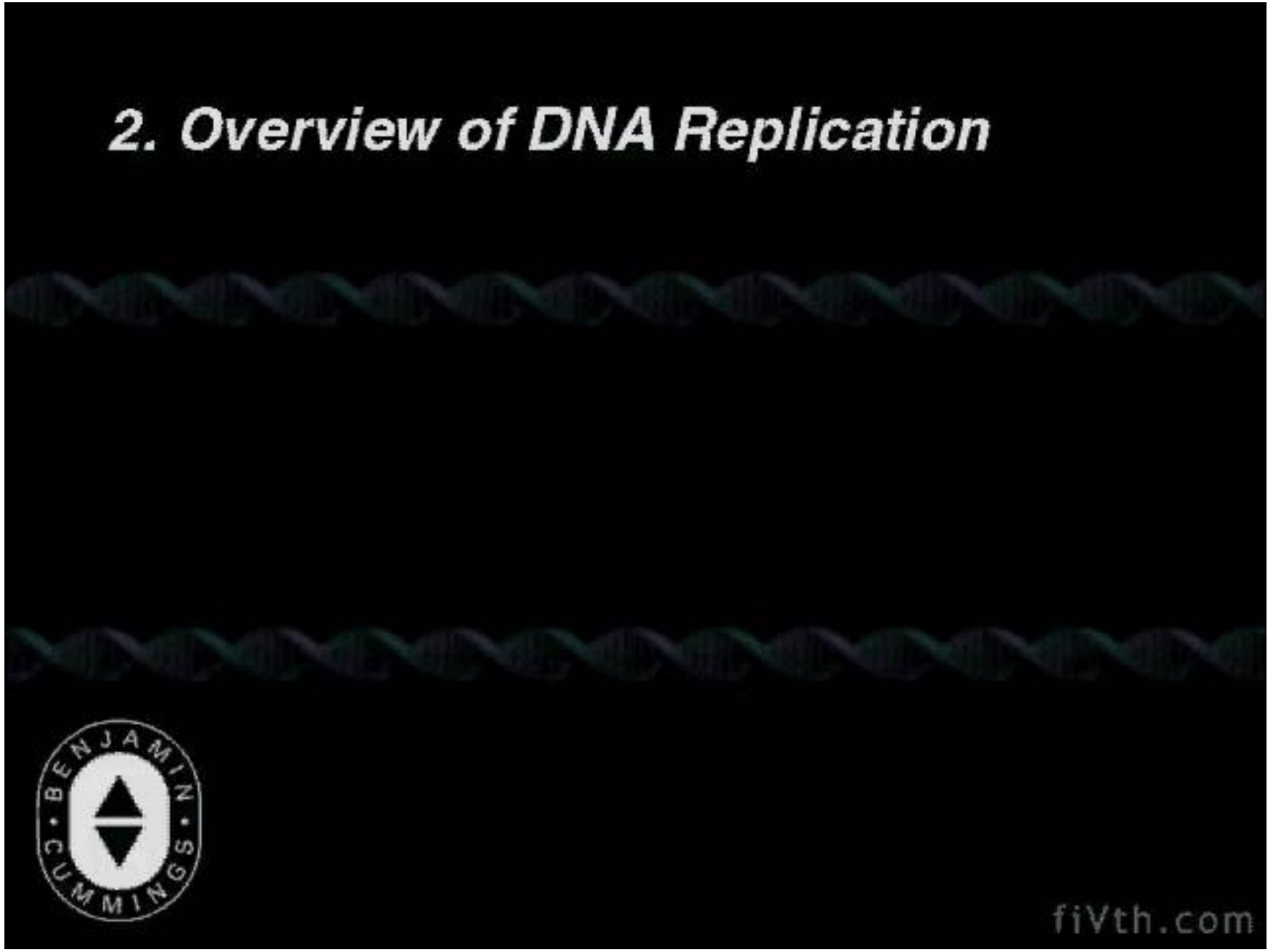
This means that in each new double helix of DNA, one strand was from the parent.



Bacterial DNA	Human DNA
• DNA length ~ 10^6 bp • Circular DNA • Replication time ~ 20 minutes • Speed of replication ~ 1000bp/s	• DNA length ~ 10^9 bp • Linear DNA • Replication time ~ 8 hours • Speed of replication ~ 50 bp/s


PROKARYOTIC
Pre-replication complex
Replication fork
ori
ori
ori
ori
REPLICATION IS BIDIRECTIONAL


EUKARYOTIC
ori
ori
ori
Initiation of replication
LIFE 10e, Figure 13.12
© 2014 Sinauer Associates, Inc.


2. Overview of DNA Replication
BENJAMIN CUMMINGS
fiVth.com

Scan the QR or use link to join

https://forms.cloud.microsoft/r/2EUqtnLUbw

In the image shown, which section do you think wi
fro DNA replication?

A - End of chromosome

B - AT-rich region

C - Region mith mixed AT/GC

D - GC-rich region

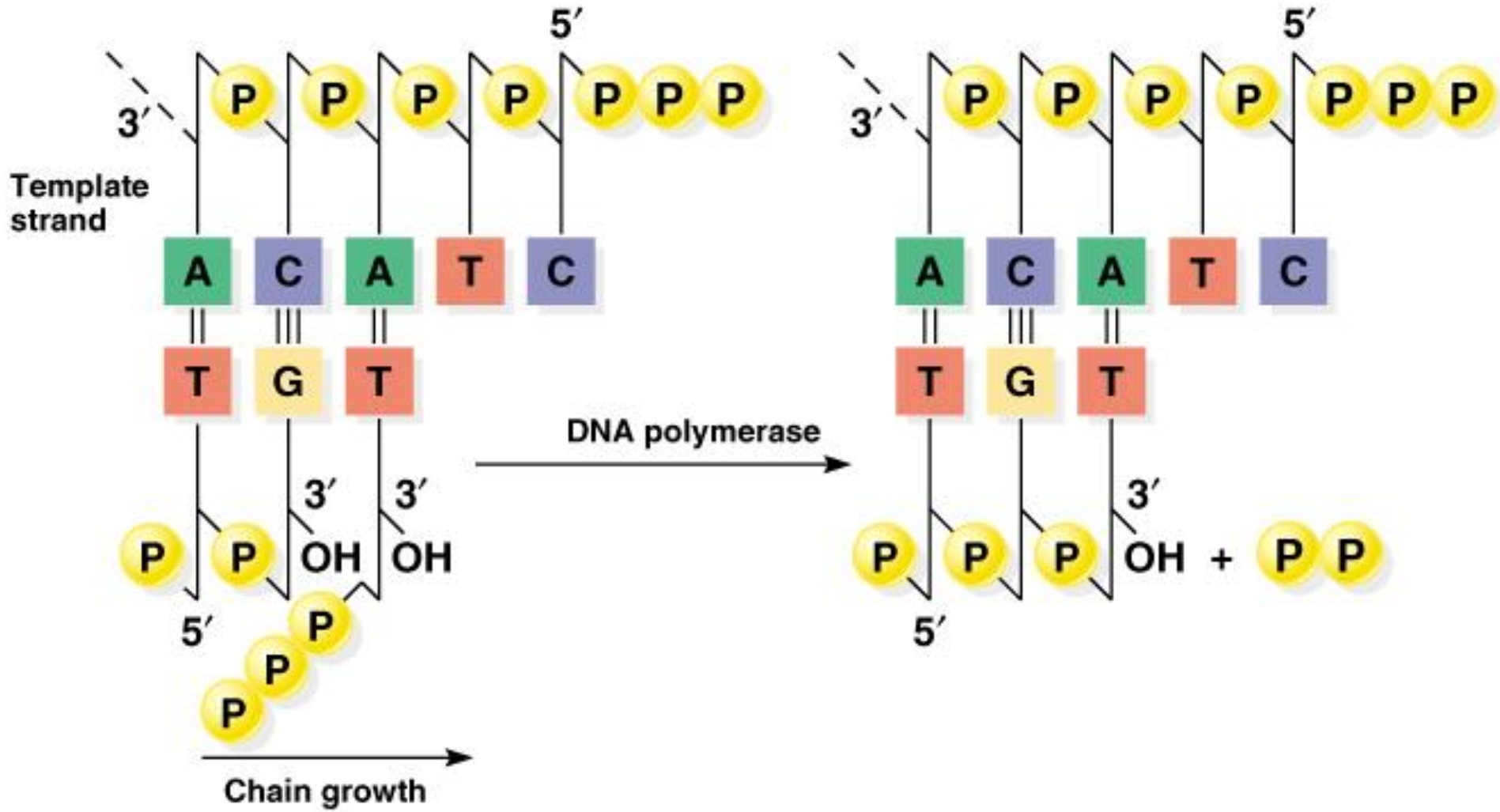
DNA Replication steps

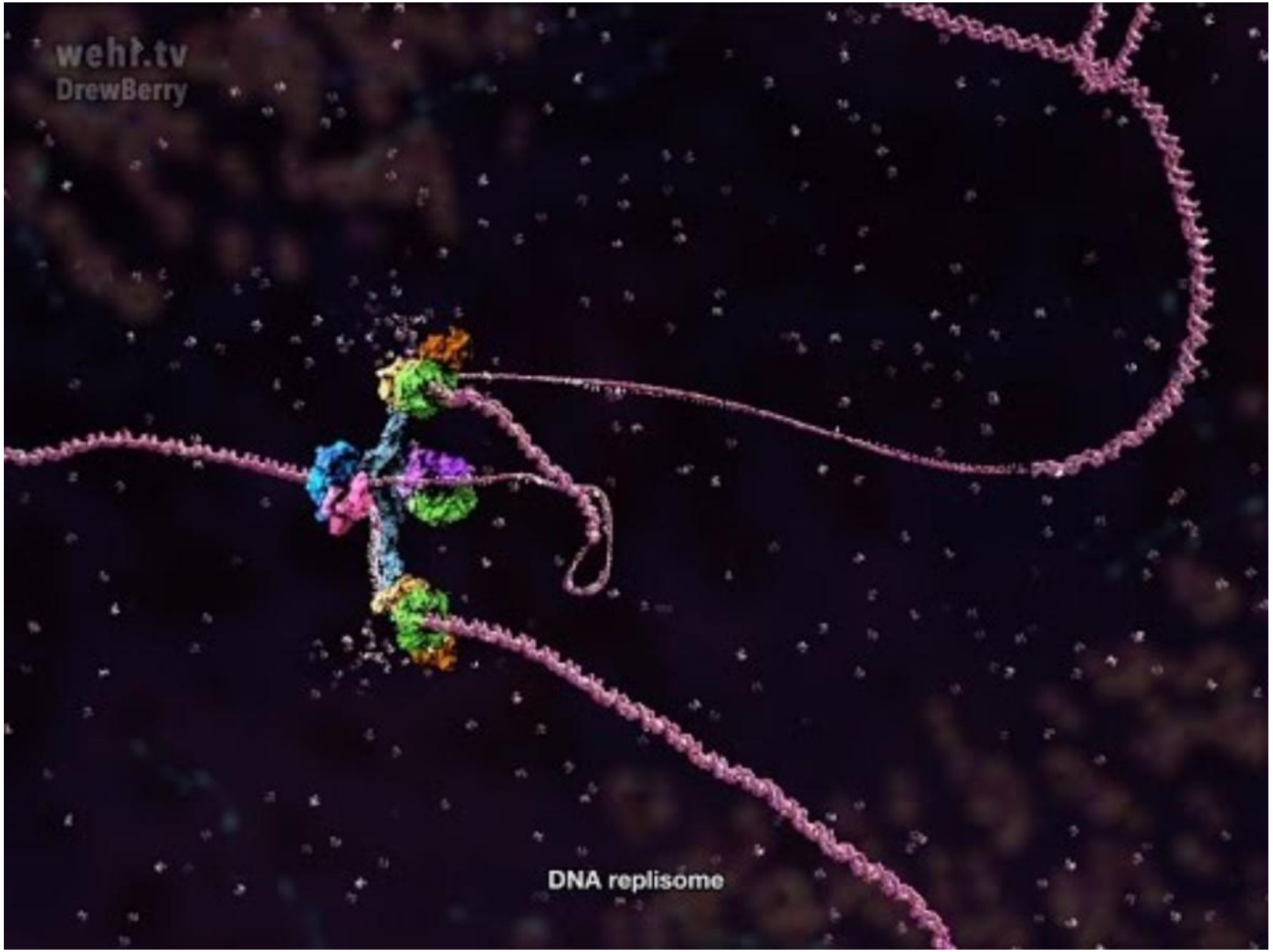
1. DNA polymerase catalyzes formation of phosphodiester bond between 3’-OH of the deoxyribose (on the last nucleotide) and the 5’-phosphate of the dNTP.

- Energy for this reaction is derived from the release of two of the three phosphates of the dNTP.

2. DNA polymerase “finds” the correct complementary dNTP at each step in the lengthening process.

3. Direction of synthesis is 5’ to 3’


5′
3′
P P P P P P P
Template strand
A C A T C
T G T
3′ 3′
P P OH OH
5′
P
P
P
Chain growth
DNA polymerase
5′
3′
P P P P P P P
A C A T C
T G T
3′
P P P OH + P P
5′



https://www.youtube.com/watch?v=sKe3UgH1AKg

DNA Replication Ingredients

NUCLEIC ACIDS

- Double-stranded DNA (dsDNA)
- Deoxynucleotide triphosphates (dNTPs)

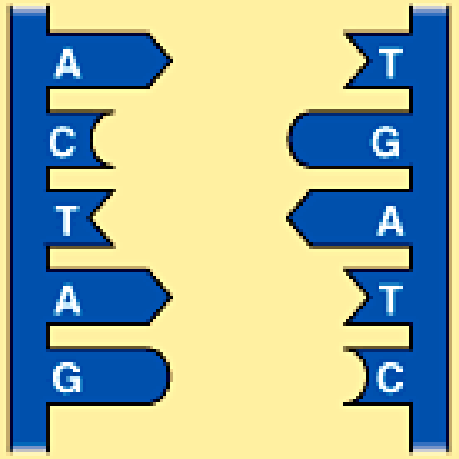
ENZYMES:

- **Helicases** – unwinds dsDNA
- **Single-stranded DNA binding (SSB) proteins**
- **Primase or a primer** – Provides starting point for Polymerising
- **DNA polymerases** – Enzyme that elongates daughter strand
- **Topoisomerase** – Relaxes DNA
- **RNAse** – Cuts RNA
- **Ligase** – Glue that binds DNA backone

DNA is elongated 5'-3'

DNA polymerase need a primer to start

Step 1: Separation of the Double Helix

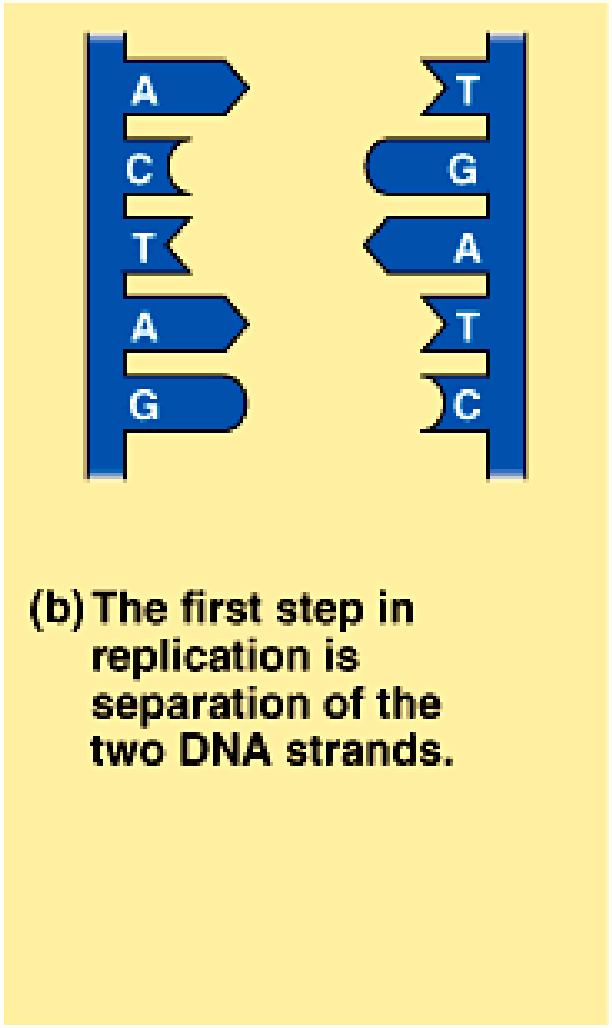


(b) The first step in replication is separation of the two DNA strands.

DNA Helicase unwinds the double helix

- The site where the enzyme begins replication is called the **origin of replication.**

Step 1: Separation of the Double Helix



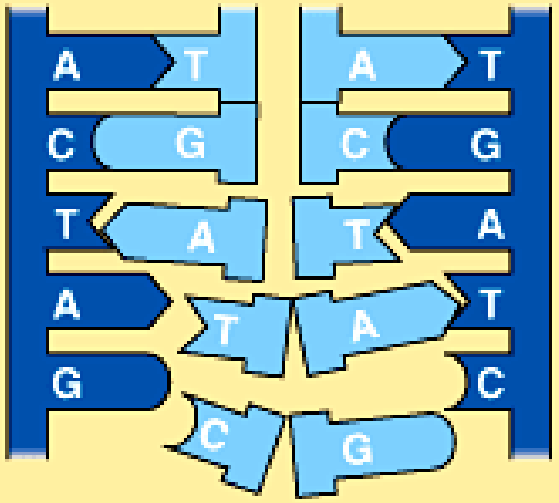
(b) The first step in replication is separation of the two DNA strands.

DNA Helicase unwinds the double helix

- The site where the enzyme begins replication is called the **origin of replication.**
- In Prokaryotes (Bacteria), there is a single origin; in eukaryotes, there are thousands

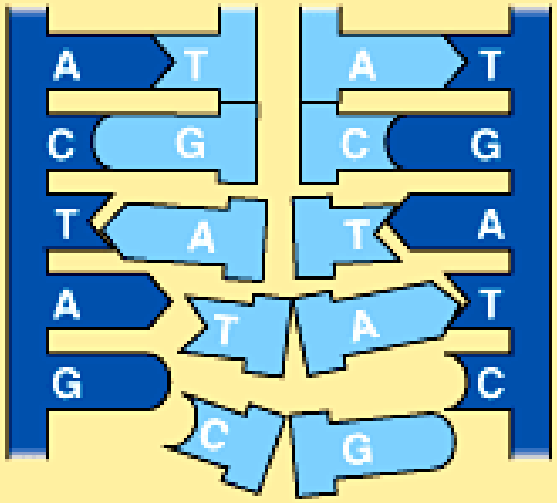
Step 2: Synthesis of complements

Enzyme- DNA polymerase



(c) Each parental strand now serves as a template that determines the order of nucleotides along a new complementary strand.

Step 2: Synthesis of complements

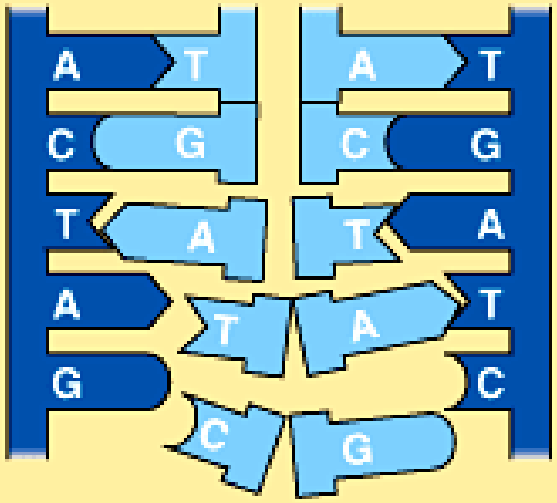


(c) Each parental strand now serves as a template that determines the order of nucleotides along a new complementary strand.

Enzyme- DNA polymerase

- DNA polymerase allows the new nucelotides to be added to the growing end of the strand one by one

Step 2: Synthesis of complements

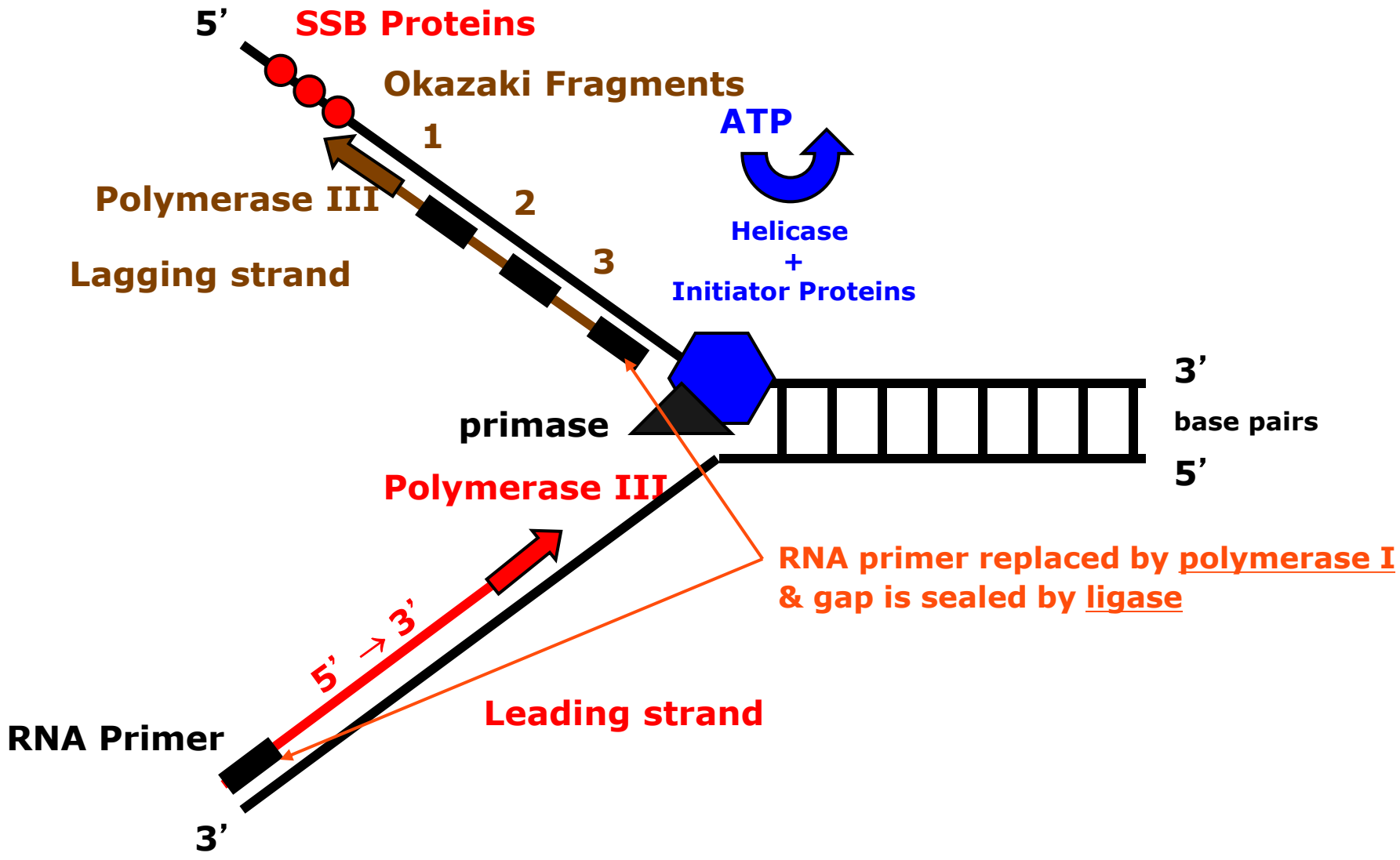


(c) Each parental strand now serves as a template that determines the order of nucleotides along a new complementary strand.

Enzyme- DNA polymerase

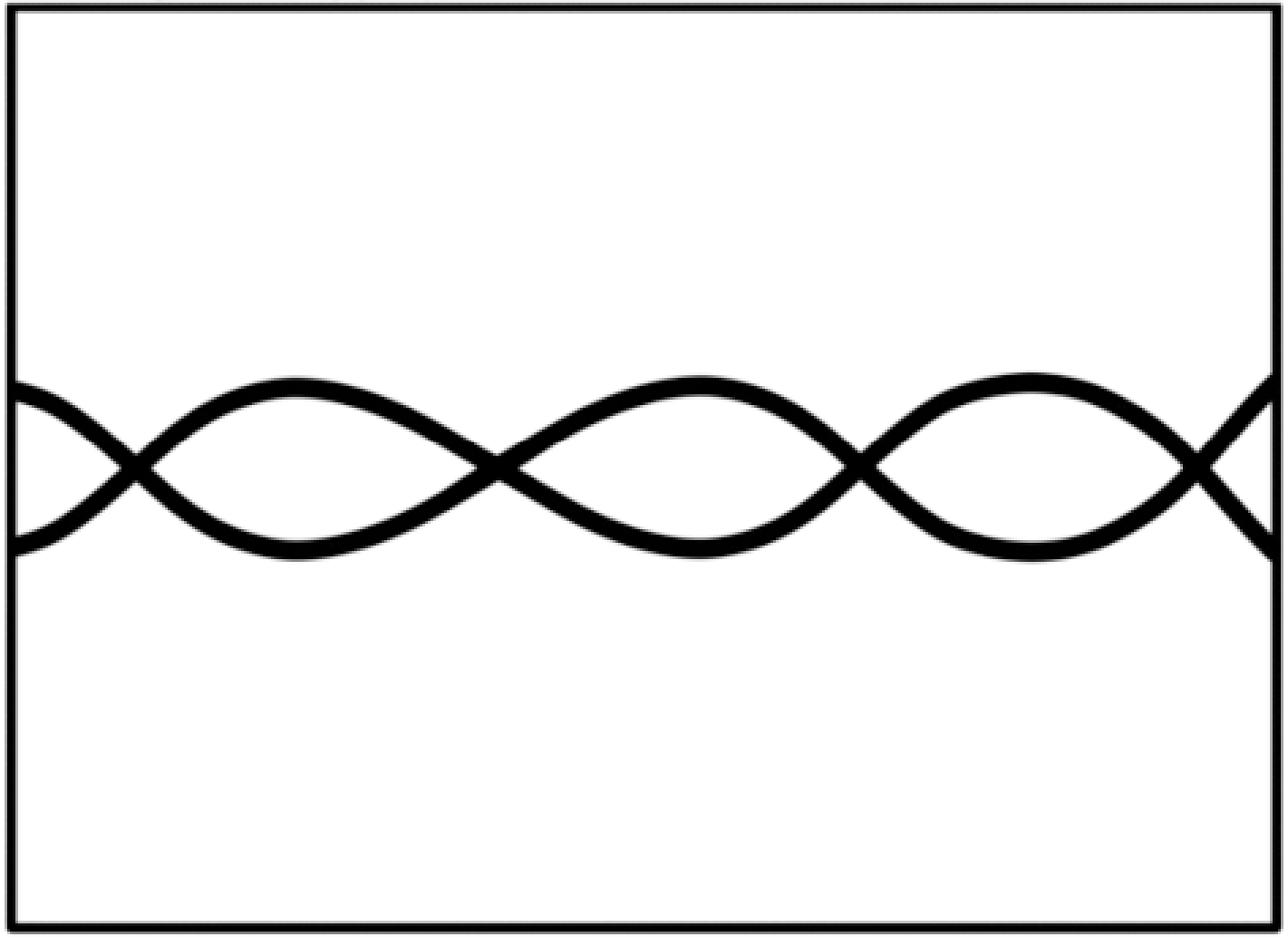
- DNA polymerase allows the new nucelotides to be added to the growing end of the strand one by one

- NOTE: This new synthesis always occurs in the same direction, 5'→3'


5’
SSB Proteins
Okazaki Fragments
1
ATP
Polymerase III
2
Helicase
+
Initiator Proteins
Lagging strand
3
3’
primase
base pairs
5’
Polymerase III
RNA primer replaced by polymerase I
& gap is sealed by ligase
5’ → 3’
Leading strand
RNA Primer
3’

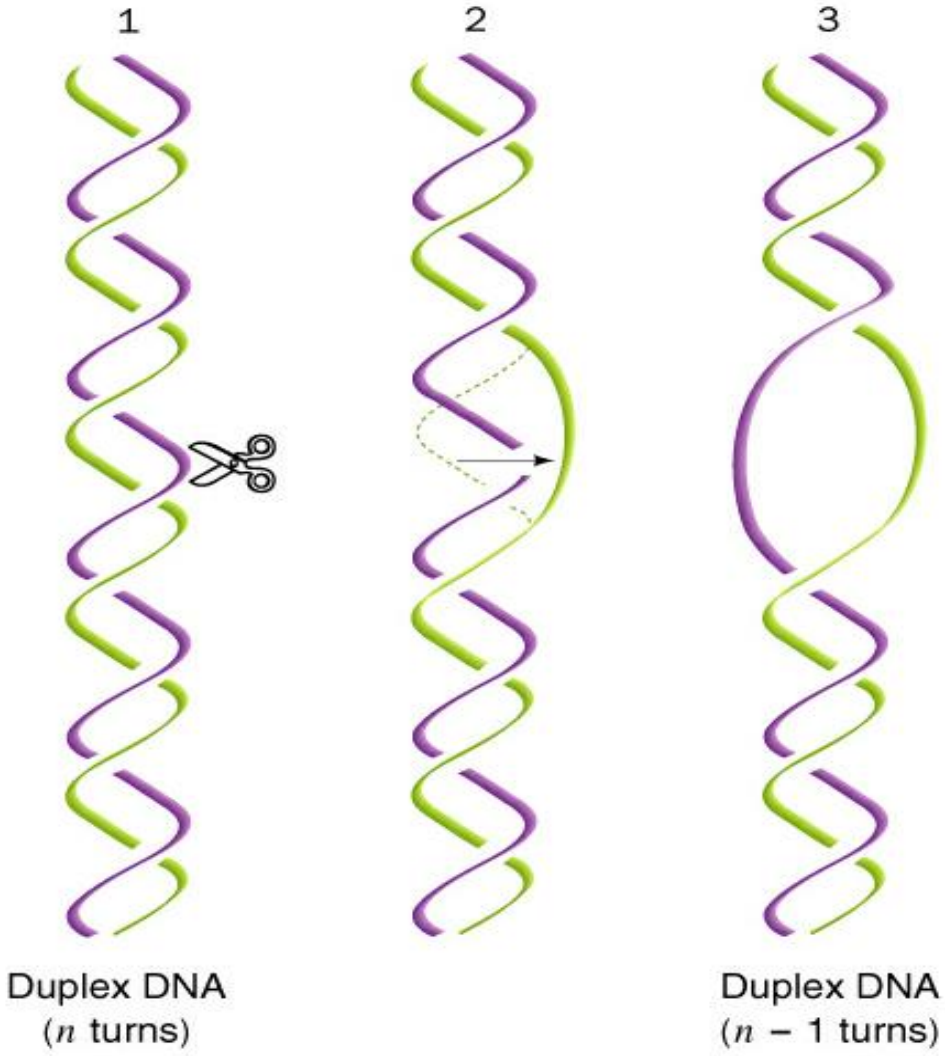
Leading and Lagging Strand in DNA Replication

@AmoebaSisters

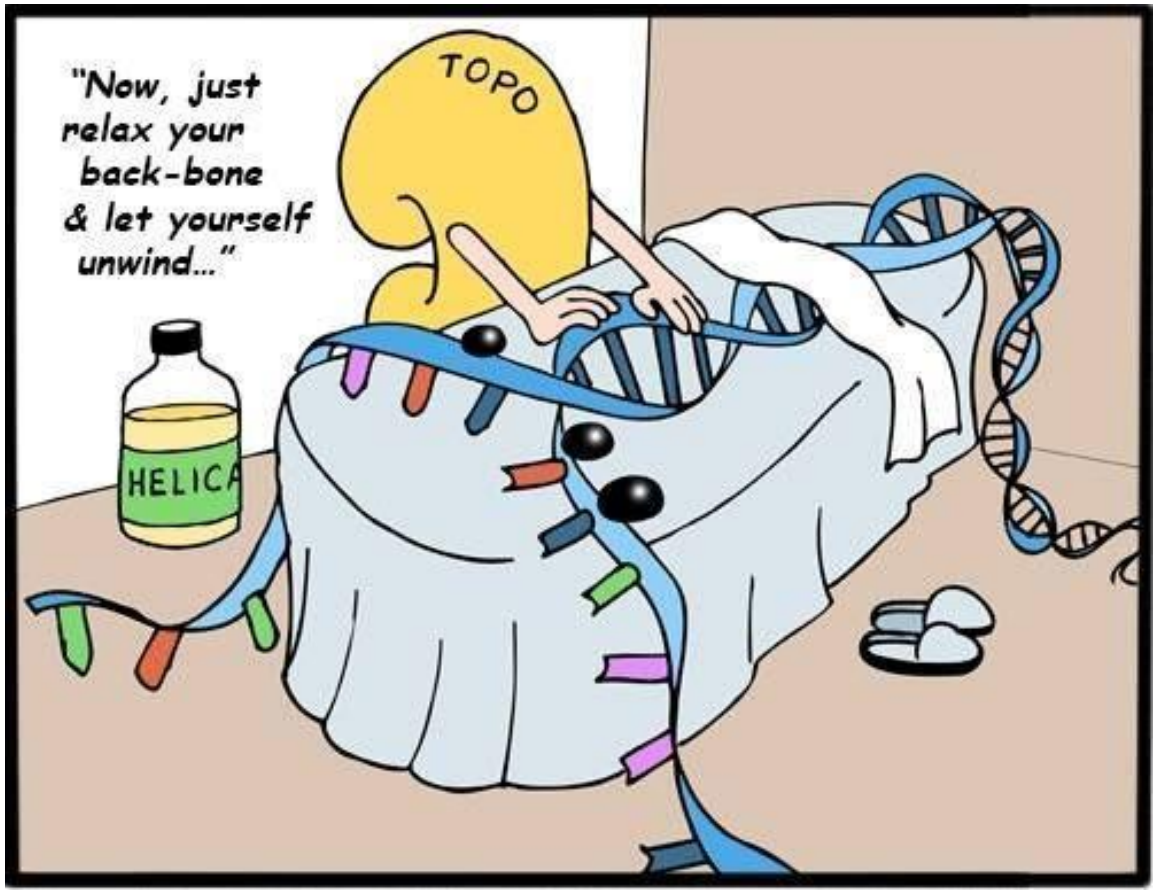


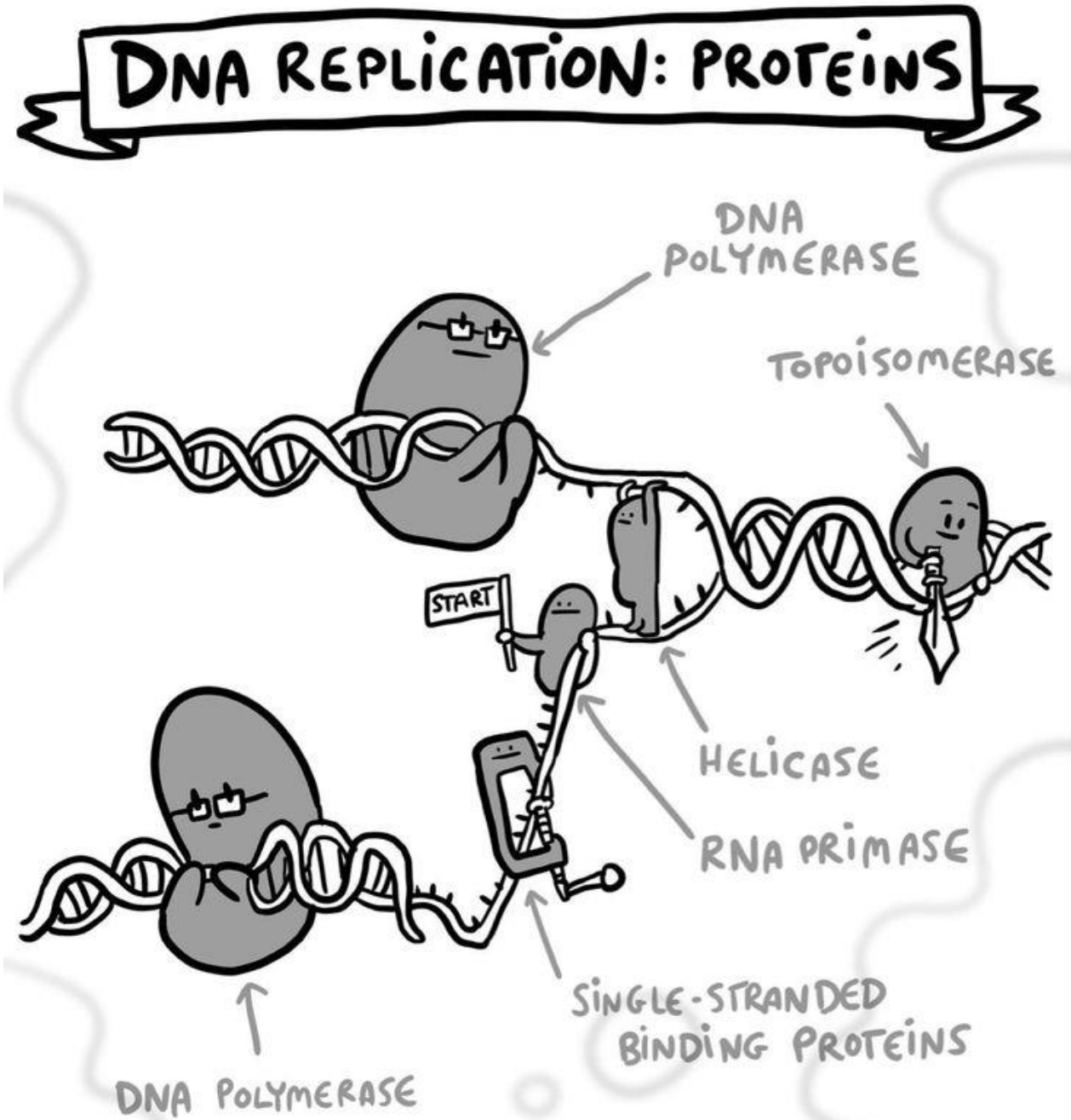


Topoisomerase

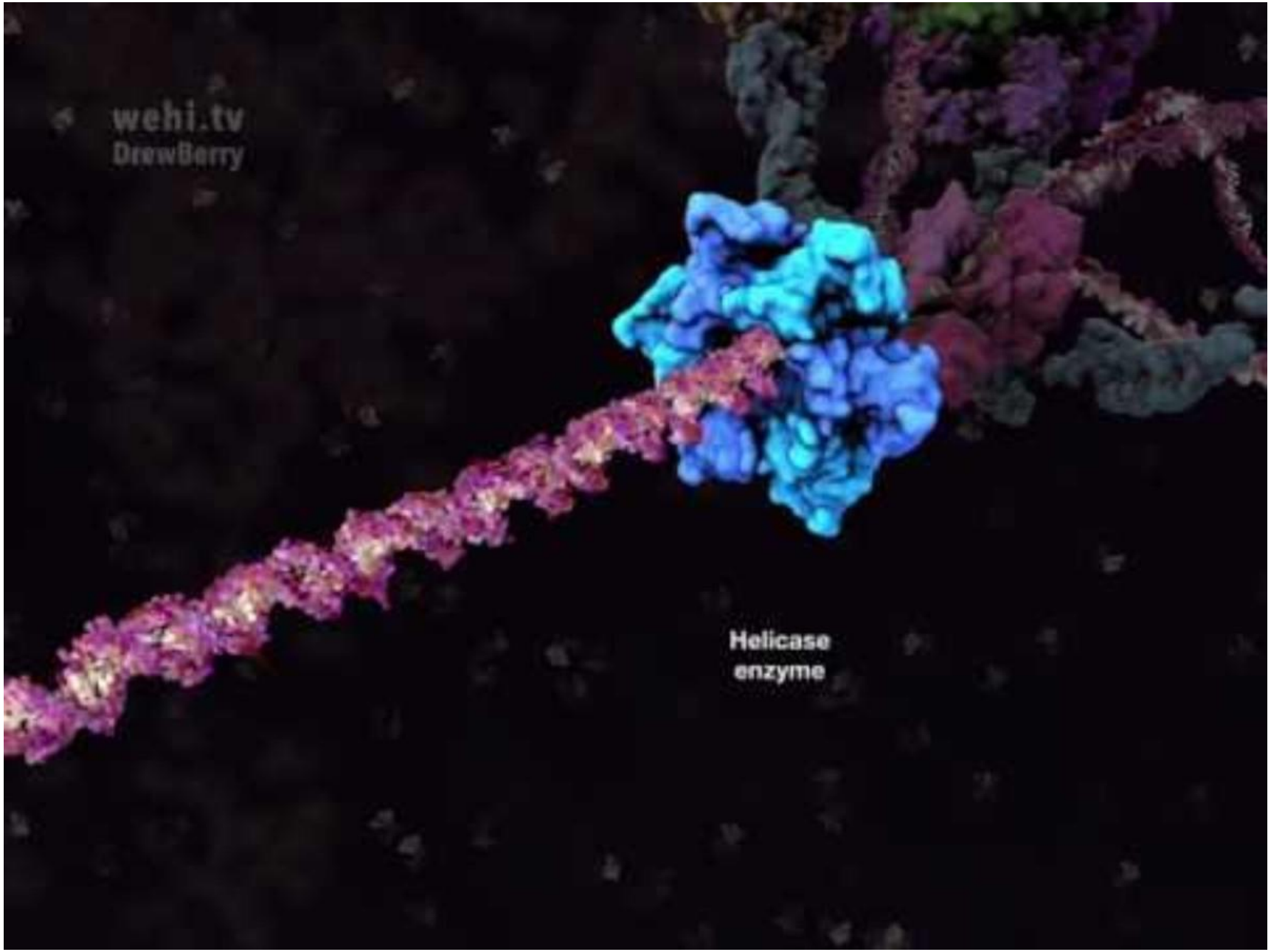


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https://www.youtube.com/watch?v=6j8CV3droDw